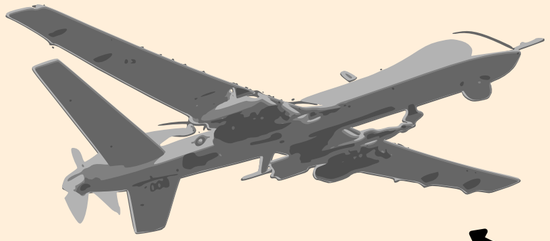
GROUND CONTROL STATION SOFTWARE



GAURAV SHAHI



WHAT IS GCS ?

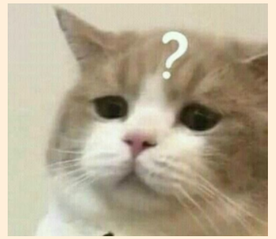


A Ground Control Station is a software-hardware hub that monitors and commands unmanned systems. It displays real-time telemetry, logs mission data, and enables remote control for aerial, marine, or land drones.



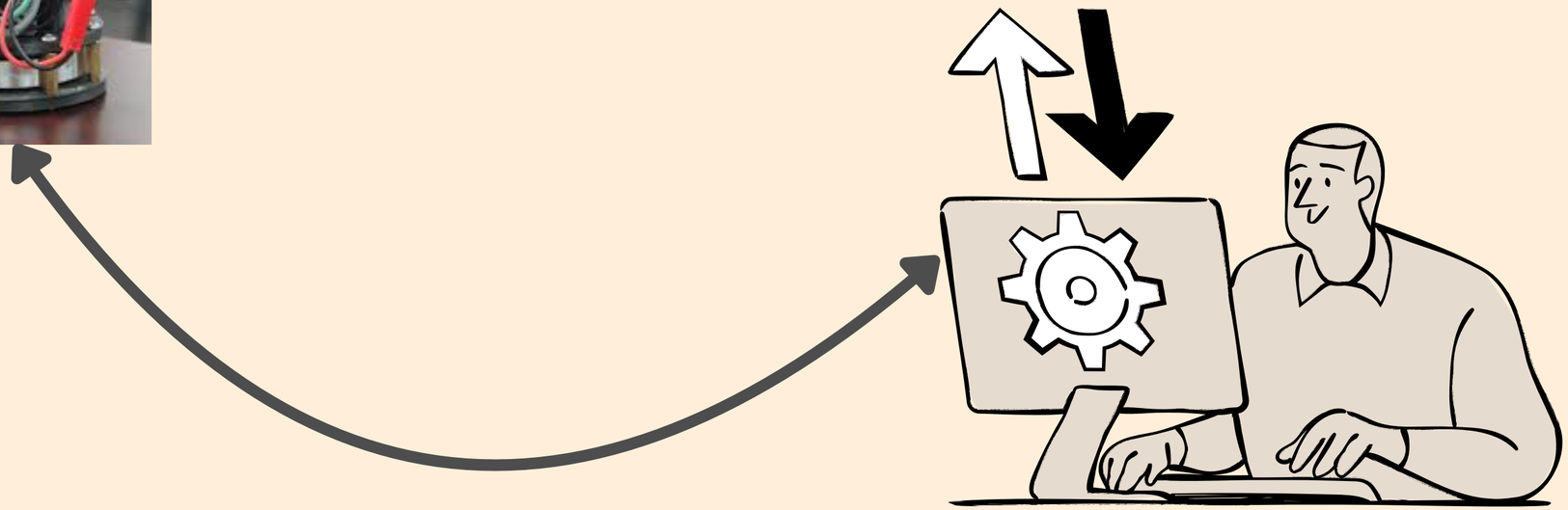


HOW DOES IT WORK ?



The GCS connects via radio, cellular, or satellite link to the payload. It decodes incoming telemetry packets, interprets sensor readings, and transmits control commands back to the vehicle’s onboard flight computer





USER INTERFACE DESIGN

Effective UI presents maps, graphs, and status indicators. Use modular widgets, clear color coding, and responsive layouts. Prioritize low latency, intuitive controls, and customizable dashboards for diverse operational needs.



PROCESS FLOW



Wireless Transmission via XBee

Data Reception on GCS (Python Serial Read)

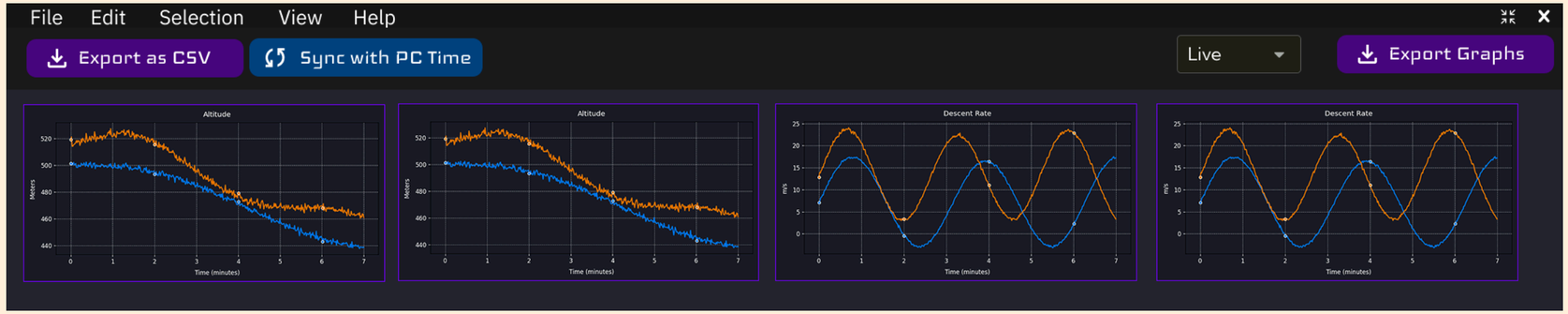


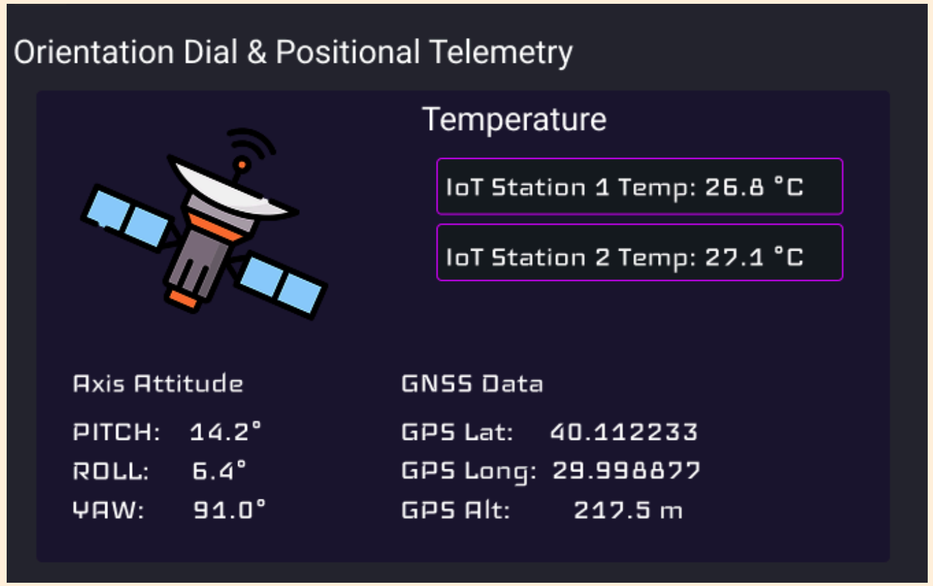
Sensor Data Collection (On Payload)

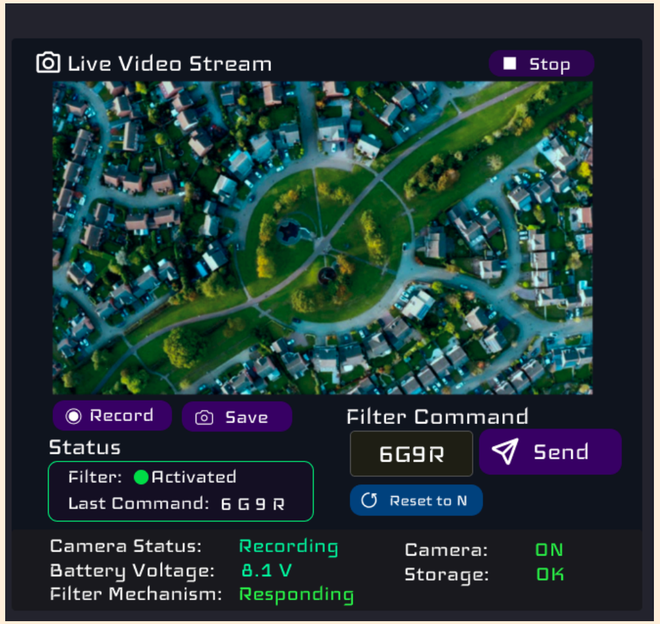
Python ↔ Electron Communication (IPC/WebSocket)

Real-Time Visualization in Electron GUI

EXAMPLES


File
Edit
Selection
View
Help
Export as CSV
Sync with PC Time
Live
Export Graphs
Altitude
Meters
Time (minutes)
Altitude
Meters
Time (minutes)
Descent Rate
m/s
Time (minutes)
Descent Rate
m/s
Time (minutes)


Orientation Dial & Positional Telemetry
Temperature
IoT Station 1 Temp: 26.8 °C
IoT Station 2 Temp: 27.1 °C
Axis Attitude
PITCH: 14.2°
ROLL: 6.4°
YAW: 91.0°
GNSS Data
GPS Lat: 40.112233
GPS Long: 29.998877
GPS Alt: 217.5 m


Live Video Stream
Stop
Record
Save
Filter Command
6G9R
Send
Status
Filter: Activated
Last Command: 6 G 9 R
Reset to N
Camera Status: Recording
Battery Voltage: 8.1 V
Filter Mechanism: Responding
Camera: ON
Storage: OK


THANK YOU
FOR LISTENING!
MILK
ANY QUESTIONS?
imgflip.com

YOUR TASK

Try to build a simple GUI

1. Using HTML/CSS/JS that prints dummy data of a graph
2. Package it using Electron/Tauri

